

UNIT-I: CLOUD COMPUTING FUNDAMENTALS

What is Cloud Computing

Definitions Of Cloud Computing:

Cloud computing is a model for enabling **ubiquitous** (Anywhere & Any time), **convenient**, **on-demand** network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

Wikipedia:--—Cloud computing is Web-based processing, where by shared resources, software, and information are provided to computers and other devices (such as smart phones) on demand over the Internet.¶

Forrester:--—Cloud computing: A pool of abstracted, highly scalable, and managed infrastructure capable of hosting end-customer applications and billed by consumption.

Introduction to Cloud Computing:

Cloud computing takes the technology, services, and applications that are similar to those on the Internet and turns them into a self-service utility. The use of the word —cloud makes reference to the two essential concepts:

- **Abstraction:** Cloud computing abstracts the details of system implementation from users and developers. Applications **run on physical systems** that aren't specified, data is stored in **locations that are unknown**, administration of systems is outsourced to others, and access by users is ubiquitous.
- **Virtualization:** Cloud computing **virtualizes systems by pooling and sharing resources**. Systems and storage can be provisioned as needed from a **centralized infrastructure**, costs are assessed on a **metered basis**, multi-tenancy is enabled, and resources are scalable with agility.

Cloud Architecture

The 5-4-3 principles put forth by National Institute of Standards and Technology (NIST) describe:

- 3 - Cloud Service Models
- 4 - Cloud Deployment Models
- 5 - Essential Characteristics of Cloud Computing

Cloud Deployment Models:

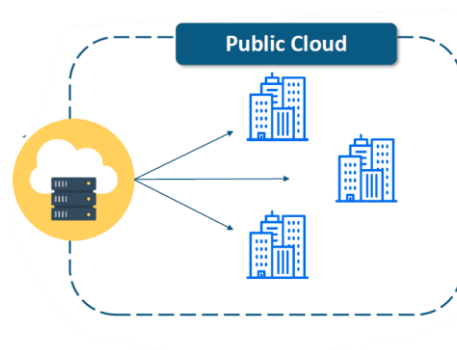
Cloud computing deployment models refer to how cloud services are made available to users and organizations. These models define the ownership, access, and control over the infrastructure that supports the cloud services. There are three main cloud computing deployment models: Public Cloud, Private Cloud, Hybrid Cloud, and Community Cloud.

Public Cloud:

Definition: In a public cloud deployment model, cloud services are provided by third-party service providers over the internet. These services are available to the general public, and multiple organizations may share the same resources and infrastructure.

Key Characteristics:

- Cost-effective as resources are shared among multiple users.
- Scalable, allowing users to increase or decrease resources as needed.
- Service providers manage and maintain the infrastructure.
- Examples of public cloud providers include Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP).



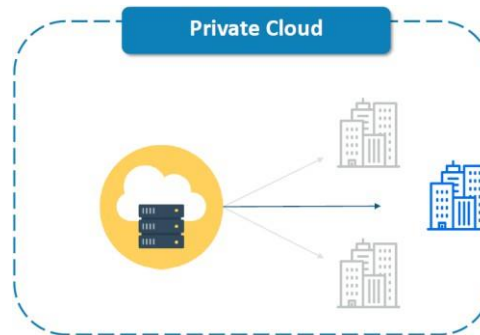
Private Cloud:

Definition: In a private cloud deployment model, cloud services are used exclusively by a single organization. The infrastructure can be located on-premises or hosted by a third-party provider, but it is dedicated to the specific needs of one organization.

Key Characteristics:

- Offers greater control and customization over the infrastructure.
- Enhanced security and privacy, as resources are not shared with other organizations.
- Suitable for organizations with specific compliance requirements or sensitive data.

- Requires more significant upfront investment and ongoing maintenance.
- Examples of private cloud solutions include VMware Cloud Foundation and OpenStack.

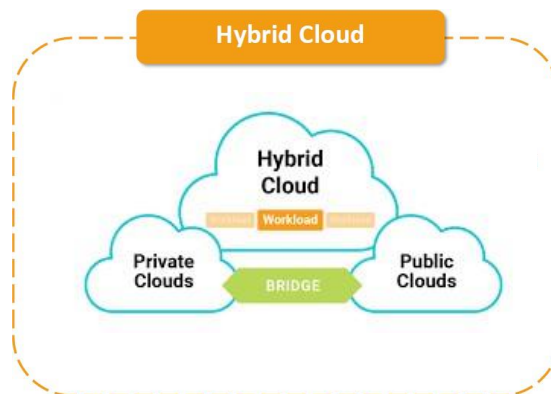


Hybrid Cloud:

Definition: The hybrid cloud deployment model combines elements of both public and private clouds. It allows data and applications to be shared between them, providing greater flexibility and more deployment options.

Key Characteristics:

- Offers a balance between the scalability of the public cloud and the control of the private cloud.
- Enables data and applications to move between public and private environments as needed.
- Useful for organizations with varying workloads or specific regulatory requirements.
- Requires effective management of data integration, security, and compliance.
- Examples of hybrid cloud solutions include AWS Outposts, Azure Hybrid Cloud, and Google Anthos.



Cloud computing Service Models:

Cloud service models, also known as cloud computing service models, refer to the different types of services that are offered through cloud computing platforms. These models define the level of control and responsibility that users have over their computing resources and services. There are three main cloud service models: Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). Let's delve into each of them:

Infrastructure as a Service (IaaS):

Definition: IaaS is the most basic cloud service model, providing virtualized computing resources over the internet. It allows users to rent virtual machines, storage, and networking components on a pay-as-you-go basis.

Key Characteristics:

- Users have control over the operating system, applications, and development frameworks.
- Offers scalability, allowing users to adjust resources based on their needs.
- Examples of IaaS providers include Amazon Web Services (AWS) EC2, Microsoft Azure Virtual Machines, and Google Compute Engine.

Platform as a Service (PaaS):

Definition: PaaS provides a platform that includes not only the infrastructure but also the tools and services needed for application development. It abstracts away the complexity of managing underlying infrastructure.

Key Characteristics:

- Users focus on developing and deploying applications without managing the underlying infrastructure.
- Offers tools for application development, such as databases, middleware, and development frameworks.
- Well-suited for developers and teams working on application development and deployment.
- Examples of PaaS providers include Heroku, Google App Engine, and Microsoft Azure App Service.

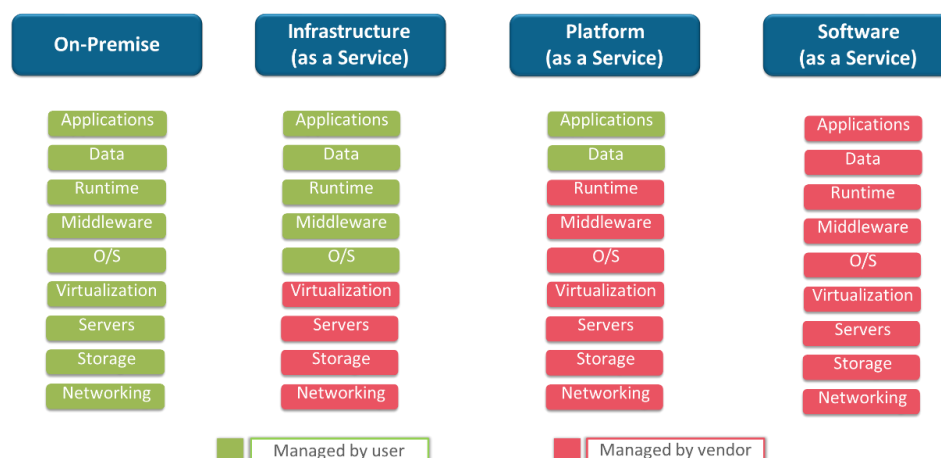
Software as a Service (SaaS):

Definition: SaaS delivers software applications over the internet on a subscription basis. Users access the software through a web browser without needing to install or maintain it locally.

Key Characteristics:

- Users have minimal control over the underlying infrastructure and application customization.
- Eliminates the need for software installation and maintenance.
- Typically charged on a subscription basis, with updates and maintenance handled by the service provider.
- Examples of SaaS applications include Salesforce, Google Workspace, and Microsoft 365.

AWS is a cloud computing provider with a wide variety of IaaS, PaaS, and SaaS offerings.



Five Essential Characteristics:

Cloud computing has five essential characteristics, which are shown in Figure 2.2. Readers can note the word *essential*, which means that if any of these characteristics is missing, then it is not cloud computing:

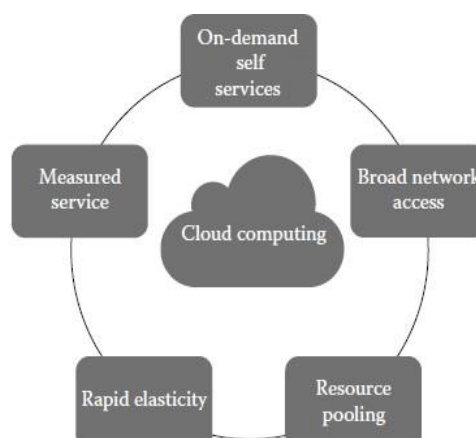


FIGURE 2.2
The essential characteristics of cloud computing.

1. **On-demand self-service:** A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service's provider.
2. **Broad network access:** Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, laptops, and personal digital assistants [PDAs])
3. **Elastic resource pooling:** The provider's computing resources are pooled to serve multiple consumers using a multitenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand. There is a sense of location independence in that the customer generally has no control or knowledge over the exact location of the provided resources but may be able to specify the location at a higher level of abstraction (e.g., country, state, or data center). Examples of resources include storage, processing, memory, and network bandwidth.
4. **Rapid elasticity:** Capabilities can be rapidly and elastically provisioned, in some cases automatically, to quickly scale out and rapidly released to quickly scale in. To the consumer, the capabilities available for provisioning often appear to be unlimited and can be purchased in any quantity at any time.
5. **Measured service:** Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts). Resource usage can be monitored, controlled, and reported providing transparency for both the provider and consumer of the utilized service.

Infrastructure as service (IaaS)

Infrastructure as a Service (IaaS) is a cloud computing model that provides virtualized computing resources over the internet. It is one of the three main categories of cloud services, alongside Platform as a Service (PaaS) and Software as a Service (SaaS). In the IaaS model, a third-party provider hosts hardware, software, servers, storage, and other infrastructure components on behalf of its users. Customers can rent these resources on a pay-as-you-go basis, scaling up or down as needed.

IaaS eliminates the need for organizations to invest in and maintain physical hardware, offering flexibility, scalability, and cost-efficiency. It is particularly useful for businesses that need to manage dynamic workloads, develop and test applications, or host websites.

The key components of IaaS:

IaaS providers use virtualization technology to create virtual instances of physical hardware resources. This allows multiple users to share the same physical hardware while maintaining isolation and security between different users' virtual environments. Users can create virtual machines (VMs) with specific configurations (CPU, RAM, storage, etc.) based on their requirements.

Compute: IaaS providers offer scalable compute resources in the form of virtual machines or containers. Users can choose the type and size of virtual machines based on their processing power, memory, and storage needs. These virtual machines can be provisioned, managed, and scaled up or down dynamically, allowing users to adapt to changing workloads and demands.

Storage: IaaS platforms provide scalable storage solutions that can accommodate various types of data, including structured, semi-structured, and unstructured data. Users can store files, databases, and other data types in the cloud, with options for block storage, object storage, and file storage. Storage resources can be scaled up or down as needed, and data is typically replicated across multiple locations for redundancy and fault tolerance.

Networking: IaaS providers offer networking services to connect virtual machines, storage, and other resources within the cloud environment. Users can set up virtual networks, subnets, and firewalls to control traffic and ensure security. Additionally, IaaS platforms often include load balancers, DNS services, and other networking features to optimize performance and availability.

Security: Security is a critical aspect of IaaS, and providers implement various measures to protect data and resources. This includes encryption for data in transit and at rest, identity and access management (IAM) for controlling user access to resources, network security controls such as firewalls and intrusion detection systems (IDS), and compliance certifications to meet regulatory requirements.

Load balancers: Load Balancing as a Service may provide load balancing capability at the infrastructure layer.

Characteristics of IaaS

1. **On-Demand Self-Service:** Users can provision computing resources (e.g., virtual machines, storage, and networking) as needed without requiring human intervention from the service provider.
2. **Scalability and Elasticity:** Resources can be scaled up or down automatically based on demand, ensuring optimal performance and cost-efficiency.
3. **Resource Pooling:** Multiple users share the same physical infrastructure, but their virtual environments are isolated and secure.
4. **Pay-as-You-Go Pricing:** Customers pay only for the resources they use, typically on an hourly, weekly, or monthly basis.
5. **Broad Network Access:** Services are accessible over the internet from a variety of devices, such as laptops, tablets, and smartphones.
6. **Virtualization:** IaaS relies heavily on virtualization technology to create virtual machines (VMs), storage, and networks that mimic physical infrastructure.
7. **Automation and Management Tools:** Providers offer tools for automating tasks like load balancing, monitoring, and resource allocation.
8. **Security and Compliance:** IaaS providers often offer built-in security features, such as firewalls, encryption, and compliance with industry standards (e.g., GDPR, HIPAA).

Suitability of IaaS

IaaS is ideal for the following use cases:

1. **Startups and Small Businesses:** Startups with limited capital can avoid the high upfront costs of purchasing and maintaining physical infrastructure.
2. **Testing and Development Environments:** IaaS provides a flexible and cost-effective environment for developing and testing applications.
3. **Web Hosting:** Businesses can host websites on IaaS platforms, benefiting from scalability and reliability.

4. **Big Data Analysis:** IaaS can handle large-scale data processing and storage requirements for big data analytics.
5. **High-Performance Computing (HPC):** IaaS supports compute-intensive tasks like scientific research, simulations, and rendering.
6. **Disaster Recovery and Backup:** IaaS provides reliable and scalable solutions for data backup and disaster recovery
7. **Variable Workloads:** Businesses with fluctuating workloads (e.g., seasonal spikes) can scale resources up or down as needed.

Pros and Cons of IaaS:

Pros of IaaS

1. **Cost-Efficiency:** Eliminates the need for capital expenditure (CapEx) on physical hardware and reduces operational costs (OpEx).
2. **Scalability:** Resources can be scaled up or down quickly to meet changing demands.
3. **Flexibility:** Users have full control over their infrastructure, including operating systems, applications, and middleware.
4. **Focus on Core Business:** Organizations can focus on their core competencies instead of managing IT infrastructure.
5. **Disaster Recovery:** IaaS providers often offer robust disaster recovery solutions, ensuring business continuity.
6. **Global Reach:** Resources can be accessed from anywhere with an internet connection, enabling remote work and global operations.
7. **Innovation:** Businesses can experiment with new technologies and applications without significant upfront investment.

Cons of IaaS

1. **Security Concerns:** Storing sensitive data on third-party servers may raise security and compliance issues.
2. **Vendor Lock-In:** Migrating from one IaaS provider to another can be challenging due to differences in platforms and APIs.

3. **Downtime:** Although rare, outages or downtime on the provider's end can disrupt operations.
4. **Limited Control:** While users have control over virtual machines, they have no control over the underlying physical infrastructure.
5. **Cost Management:** Without proper monitoring, costs can spiral out of control due to over-provisioning or inefficient resource usage.
6. **Technical Expertise Required:** Managing IaaS resources requires technical knowledge, which may necessitate hiring skilled personnel.

Summary of IaaS providers

1. Amazon Web Services (AWS)

- **Overview:** AWS is the most widely used IaaS provider, offering a comprehensive suite of cloud services.
- **Key Services:**
 - **Amazon EC2 (Elastic Compute Cloud):** Scalable virtual servers for running applications.
 - **Amazon S3 (Simple Storage Service):** Object storage for data backup, archiving, and analytics.
 - **Amazon VPC (Virtual Private Cloud):** Isolated cloud resources for secure networking.
 - **AWS Lambda:** Serverless computing for running code without managing servers.
 - **Amazon RDS (Relational Database Service):** Managed relational databases.
 - **AWS Elastic Beanstalk:** Platform for deploying and managing applications.
 - **Amazon CloudFront:** Content delivery network (CDN) for fast content delivery.
 - **AWS Backup:** Automated backup solutions for data protection.

2. Microsoft Azure

- **Overview:** Azure is a leading cloud platform that integrates seamlessly with Microsoft products and services.
- **Key Services:**
 - **Azure Virtual Machines:** Scalable virtual servers for various workloads.
 - **Azure Blob Storage:** Object storage for unstructured data.
 - **Azure Virtual Network:** Secure and isolated cloud networking.
 - **Azure Kubernetes Service (AKS):** Managed Kubernetes for containerized applications.
 - **Azure SQL Database:** Managed relational database service.
 - **Azure DevOps:** Tools for CI/CD and application development.
 - **Azure Backup:** Data backup and recovery solutions.
 - **Azure CDN:** Content delivery network for global content distribution.

3. Google Cloud Platform (GCP)

- **Overview:** GCP is known for its data analytics, machine learning, and open-source integrations.
- **Key Services:**
 - **Google Compute Engine:** Scalable virtual machines for various workloads.
 - **Google Cloud Storage:** Object storage for data storage and retrieval.
 - **Google Kubernetes Engine (GKE):** Managed Kubernetes service.
 - **BigQuery:** Serverless data warehouse for analytics.
 - **Cloud SQL:** Managed relational database service.
 - **Cloud Functions:** Serverless computing for event-driven applications.
 - **Cloud CDN:** Content delivery network for fast content delivery.
 - **Cloud Storage Transfer Service:** Data migration and transfer solutions.

Platform as a Service

Platform as a Service (PaaS) is a cloud computing model that provides a platform and environment for developers to build, deploy, and manage applications without worrying about the underlying infrastructure. PaaS abstracts the complexities of hardware, operating systems, and middleware, allowing developers to focus solely on coding and innovation. It is one of the three main cloud service models, alongside Infrastructure as a Service (IaaS) and Software as a Service (SaaS).

PaaS is particularly useful for developers, startups, and enterprises looking to streamline application development, reduce time-to-market, and minimize operational overhead.

2. Key Components of PaaS:

1. **Programming languages:** PaaS providers provide a wide variety of programming languages for the developers to develop applications. Some of the popular programming languages provided by PaaS vendors are Java, Perl, PHP, Python, Ruby, Scala, Clojure and so on.

2. **Application frameworks:** PaaS vendors provide application frameworks that simplify the application development. Some of the popular application development frameworks provided by a PaaS provider include Node.js, Rails, Drupal, Joomla, WordPress, Django, EE6, Spring, Play, Sinatra, Rack, and Zend.

3. **Database:** Since every application needs to communicate with the databases, it becomes a must-have tool for every application. PaaS providers are providing databases also with their PaaS platforms. The popular databases provided by the popular PaaS vendors are ClearDB, PostgreSQL, Cloudant, Membase, MongoDB, and Redis.

4. **Other tools:** PaaS providers provide all the tools that are required to develop, test, and deploy an application.

Characteristics of PaaS

1. **Development Tools:** PaaS provides integrated development environments (IDEs), libraries, frameworks, and other tools to facilitate application development.
2. **Middleware:** Includes operating systems, runtime environments, and databases, which are managed by the provider.

3. **Automated Deployment:** PaaS platforms often include tools for automating the deployment of applications.
4. **Scalability:** Applications can scale automatically based on demand, ensuring optimal performance.
5. **Multi-Tenancy:** Multiple users can access the same development tools and infrastructure, with their environments isolated and secure.
6. **Collaboration:** PaaS enables teams to collaborate on projects in real-time, regardless of their location.
7. **Integration:** PaaS platforms often integrate with other cloud services, APIs, and third-party tools.
8. **Security and Compliance:** PaaS providers offer built-in security features, such as encryption, firewalls, and compliance with industry standards (e.g., GDPR, HIPAA).

Suitability of PaaS

PaaS is ideal for the following use cases:

1. **Application Development:** Developers can use PaaS to build, test, and deploy applications quickly and efficiently.
2. **API Development and Management:** PaaS provides tools for creating, deploying, and managing APIs.
3. **Internet of Things (IoT):** PaaS platforms can handle the data processing and storage requirements of IoT applications.
4. **Business Analytics:** PaaS offers tools for data analysis, visualization, and reporting.
5. **DevOps:** PaaS supports continuous integration and continuous deployment (CI/CD) pipelines, enabling faster and more reliable software delivery.
6. **Database Management:** PaaS provides managed database services, eliminating the need for manual database administration.
7. **Hybrid Cloud Development:** PaaS can be used to develop applications that run across both on-premises and cloud environments.

Pros and Cons of PaaS,

Pros of PaaS

1. **Cost-Efficiency:** Reduces the need for investing in and maintaining physical infrastructure.
2. **Faster Time-to-Market:** Developers can focus on coding, speeding up the development process.
3. **Scalability:** Applications can scale automatically to handle increased demand.
4. **Ease of Use:** Pre-configured environments and tools simplify the development process.
5. **Collaboration:** Enables real-time collaboration among development teams.
6. **Security:** Built-in security features and compliance with industry standards.
7. **Innovation:** Developers can experiment with new technologies and frameworks without significant upfront investment.

Cons of PaaS

1. **Vendor Lock-In:** Migrating applications from one PaaS provider to another can be challenging due to differences in platforms and APIs.
2. **Limited Control:** Developers have no control over the underlying infrastructure, which may limit customization options.
3. **Security Concerns:** Storing sensitive data on third-party platforms may raise security and compliance issues.
4. **Downtime:** Outages or downtime on the provider's end can disrupt application development and deployment.
5. **Cost Management:** Without proper monitoring, costs can escalate due to over-provisioning or inefficient resource usage.
6. **Technical Expertise Required:** While PaaS simplifies many aspects of development, it still requires technical knowledge to use effectively.

Summary of PaaS providers

Several leading companies offer PaaS solutions, each with its own strengths and features:

1. Amazon Web Services (AWS):

- **AWS Elastic Beanstalk:** Automates application deployment and management.
- **AWS Lambda:** Serverless computing for running code without managing servers.

2. Microsoft Azure:

- **Azure App Service:** Build, deploy, and scale web apps and APIs.
- **Azure Functions:** Serverless computing for event-driven applications.

3. Google Cloud Platform (GCP):

- **Google App Engine:** Fully managed platform for building and deploying applications.
- **Google Cloud Functions:** Serverless computing for running code in response to events.

4. IBM Cloud:

- **IBM Cloud Foundry:** Open-source PaaS for building and deploying applications.
- **IBM Watson:** AI and machine learning services integrated into the PaaS.

5. Oracle Cloud:

- **Oracle Cloud Platform:** Comprehensive PaaS offering for application development, integration, and analytics.
- **Oracle Autonomous Database:** Self-driving, self-securing, and self-repairing database.

6. Salesforce:

- **Salesforce Platform (Force.com):** PaaS for building enterprise applications on the Salesforce ecosystem.

Software as a service

Software as a Service (SaaS) is a cloud computing model that delivers software applications over the internet on a subscription basis. Instead of installing and maintaining software on individual devices, users access the application via a web browser. SaaS eliminates the need for complex hardware and software management, as the provider handles everything from infrastructure to updates and security. It is one of the three main cloud service models, alongside **Infrastructure as a Service (IaaS)** and **Platform as a Service (PaaS)**.

SaaS is widely used for business applications like customer relationship management (CRM), enterprise resource planning (ERP), collaboration tools, and more.

Key Components of SaaS

1. **Business services:** Most of the SaaS providers started providing a variety of business services that attract start-up companies. The business SaaS services include ERP, CRM, billing, sales, and human resources.
2. **Social networks:** Since social networking sites are extensively used by the general public, many social networking service providers adopted SaaS for their sustainability. Since the number of users of the social networking sites is increasing exponentially, cloud computing is the perfect match for handling the variable load.
3. **Document management:** Since most of the enterprises extensively use electronic documents, most of the SaaS providers started providing services that are used to create, manage, and track electronic documents.
4. **Mail services:** E-mail services are currently used by many people. The future growth in e-mail usage is unpredictable. To handle the unpredictable number of users and the load on e-mail services, most of the

Characteristics of SaaS

1. **Web-Based Access:** SaaS applications are accessed through a web browser, making them platform-independent.
2. **Subscription-Based Pricing:** Users pay a recurring fee (monthly or annually) to access the software.

3. **Automatic Updates:** The provider manages updates and patches, ensuring users always have the latest version.
4. **Multi-Tenancy:** A single instance of the application serves multiple customers (tenants), with data isolation ensured.
5. **Scalability:** SaaS applications can scale up or down based on user demand.
6. **Integration:** SaaS applications often integrate with other software and services via APIs.
7. **Security and Compliance:** Providers offer built-in security features and compliance with industry standards (e.g., GDPR, HIPAA).
8. **Accessibility:** Applications can be accessed from any device with an internet connection.

SaaS Suitability

Suitability of SaaS (Tabular Format)

Use Case	Why SaaS is Suitable	Examples of SaaS Applications
Customer Relationship Management (CRM)	Centralized platform for managing customer interactions and data.	Salesforce, HubSpot, Zoho CRM
Enterprise Resource Planning (ERP)	Streamlines business processes like finance, HR, and supply chain management.	SAP S/4HANA Cloud, Oracle ERP Cloud, NetSuite
Collaboration and Communication	Enables real-time collaboration and communication among teams.	Microsoft 365, Google Workspace, Slack
Human Resources Management (HRM)	Simplifies HR tasks like payroll, recruitment, and performance management.	Workday, BambooHR, Zoho People
Project Management	Provides tools for planning, tracking, and managing projects.	Asana, Trello, Monday.com

Use Case	Why SaaS is Suitable	Examples of SaaS Applications
Marketing Automation	Automates marketing tasks like email campaigns, social media, and analytics.	HubSpot, Marketo, Mailchimp
E-Commerce	Manages online stores, inventory, and customer orders.	Shopify, BigCommerce, WooCommerce
Accounting and Finance	Simplifies financial management, invoicing, and tax compliance.	QuickBooks Online, Xero, FreshBooks
File Storage and Sharing	Provides secure storage and sharing of files and documents.	Dropbox, Google Drive, OneDrive
Customer Support	Manages customer support tickets and interactions.	Zendesk, Freshdesk, Intercom

Pros and Cons of SaaS

Pros of SaaS

1. **Cost-Efficiency:** No upfront hardware or software costs; pay-as-you-go pricing.
2. **Ease of Use:** No installation or maintenance required; accessible via a web browser.
3. **Automatic Updates:** Providers handle updates, ensuring users always have the latest features.
4. **Scalability:** Easily scale up or down based on business needs.
5. **Accessibility:** Access applications from anywhere with an internet connection.
6. **Integration:** Seamless integration with other SaaS applications and APIs.
7. **Security:** Providers offer robust security features and compliance with industry standards.

Cons of SaaS

- 1. **Vendor Lock-In:** Migrating data and workflows to another provider can be challenging.
- 2. **Limited Customization:** SaaS applications may not offer the same level of customization as on-premise software.
- 3. **Internet Dependency:** Requires a stable internet connection for access.
- 4. **Data Security Concerns:** Storing sensitive data on third-party servers may raise security and compliance issues.
- 5. **Downtime:** Outages or downtime on the provider’s end can disrupt operations.
- 6. **Subscription Costs:**Long-term subscription fees can add up over time.

SaaS Providers (Tabular Format)

Provider	Key SaaS Offerings	Target Audience
Salesforce	CRM, Marketing Cloud, Service Cloud, Salesforce Platform	Enterprises, SMBs
Microsoft	Microsoft 365 (Office 365), Dynamics 365, Azure SaaS offerings	Enterprises, SMBs, Individuals
Google	Google Workspace (Gmail, Docs, Sheets, Drive), Google Cloud SaaS offerings	Enterprises, SMBs, Individuals
Adobe	Adobe Creative Cloud, Adobe Sign, Adobe Experience Cloud	Creative Professionals, Enterprises
Oracle	Oracle CRM, Oracle ERP Cloud, Oracle HCM Cloud	Enterprises
SAP	SAP S/4HANA Cloud, SAP SuccessFactors, SAP Customer Experience	Enterprises
Zoho	Zoho CRM, Zoho Books, Zoho Projects, Zoho Workplace	SMBs, Enterprises
Shopify	E-Commerce Platform, Shopify Payments, Shopify POS	E-Commerce Businesses